Note

The equation of tangent at P to a circle is $y = mx \pm a\sqrt{1+m^2}$.

Theorem 5.4

From any point outside the circle $x^2 + y^2 = a^2$ two tangents can be drawn.

Proof

Let $P(x_1, y_1)$ be a point outside the circle. The equation of the tangent is $y = mx \pm a\sqrt{1+m^2}$. It passes through (x_1, y_1) . Therefore

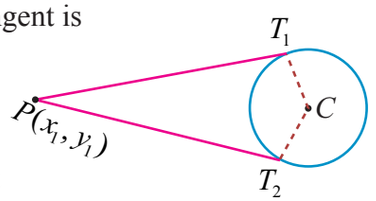


Fig.5.15

$$y_1 = mx_1 \pm a\sqrt{1+m^2}$$

$y_1 - mx_1 = a\sqrt{1+m^2}$. Squaring both sides, we get

$$(y_1 - mx_1)^2 = a^2(1+m^2)$$

$$y_1^2 + m^2x_1^2 - 2mx_1y_1 - a^2 - a^2m^2 = 0$$

$$m^2(x_1^2 - a^2) - 2mx_1y_1 + (y_1^2 - a^2) = 0 .$$

This quadratic equation in m gives two values for m .
These values give two tangents to the circle $x^2 + y^2 = a^2$.

Note

(1) If (x_1, y_1) is a point outside the circle, then both the tangents are real.

(2) If (x_1, y_1) is a point inside the circle, then both the tangents are imaginary.

(3) If (x_1, y_1) is a point on the circle, then both the tangents coincide.

Example 5.11

Find the equations of the tangent and normal to the circle $x^2 + y^2 = 25$ at $P(-3, 4)$.

Solution

Equation of tangent to the circle at $P(x_1, y_1)$ is $xx_1 + yy_1 = a^2$.

That is, $x(-3) + y(4) = 25$

$$-3x + 4y = 25$$

Equation of normal is $xy_1 - yx_1 = 0$

That is, $4x + 3y = 0$.

Example 5.12

If $y = 4x + c$ is a tangent to the circle $x^2 + y^2 = 9$, find c .

Solution

The condition for the line $y = mx + c$ to be a tangent to the circle $x^2 + y^2 = a^2$ is $c^2 = a^2(1+m^2)$.

Then, $c = \pm\sqrt{9(1+16)}$

$$c = \pm 3\sqrt{17} .$$

Example 5.13

A road bridge over an irrigation canal have two semi circular vents each with a span of $20m$ and the supporting pillars of width $2m$. Use Fig.5.16 to write the equations that represent the semi-verticular vents.

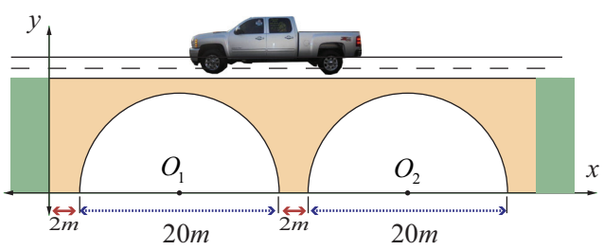


Fig.5.16

Solution

Let O_1 O_2 be the centres of the two semi circular vents.

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